KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We study convergence to non-equilibrium steady states (NESS) for dissipative evolutions on a Hilbert space generated by $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, where $\gamma > 0$ is a friction parameter, $\mathcal{R} \leq 0$ is a self-adjoint dissipator, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta \geq 0$ measures the departure from detailed balance. Our main question is: when does kinetic acceleration (lifting) survive, and how does the non-equilibrium strength η reduce the rate?

Within the Dolbeault–Mouhot–Schmeiser (DMS) hypocoercivity framework [7, 6], we build a modified Lyapunov functional and obtain an explicit η-dependent rate together with a computable stability threshold $\eta^*(\gamma)$. At the optimal friction $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$ (where $\lambda(Q_O)$ is the spectral gap of the reversible backbone Q_O in the effective generator $\mathcal{L}_O$), the threshold scales as $\eta^* \asymp \sqrt{\lambda(Q_O)}$ and the rate improves from the overdamped diffusive scale $\mathcal{O}(\lambda(Q_O)/\gamma)$ to the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$. An independent overdamped ceiling shows that this acceleration is genuinely a finite-friction effect. Therefore, systems with larger reversible gap can tolerate stronger non-equilibrium driving while keeping accelerated convergence. We also propose a simple algorithmic lift from a given $\mathcal{L}_O$ to a full generator $\mathcal{L}$. Applications are given to forced underdamped Langevin dynamics and to a boundary-driven quantum toy model.

1. Introduction

1.1. **Motivation and main question.** Many systems of interest—forced diffusions [11, 19], boundary-driven quantum chains [16], and active matter models [12]—operate away from detailed balance. Their generators contain a non-reversible component of strength η and converge to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), lifting to a larger phase space can accelerate mixing [5, 18, 8]: at optimal intermediate friction, the rate improves from the diffusive scale to its square root (quadratic speedup). In this paper we ask:

> For a system that is η away from equilibrium, under what conditions does the kinetic acceleration (lifting) persist, and how does the convergence rate degrade as a function of η?

Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is a friction (scale-separation) parameter, $\mathcal{R} \leq 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ is a non-reversible perturbation whose strength $\eta \geq 0$ is *given by the physics*. In the singular limit $\gamma \to \infty$, adiabatic elimination [14, 23] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. This overdamped limit gives a decay rate of order $\lambda(\mathcal{L}_O)/\gamma$, i.e., a diffusive bottleneck. We then ask: (i) whether the quadratic acceleration at finite friction survives the non-equilibrium perturbation, and (ii) how large η can be before the accelerated regime breaks down.

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1.2. **Main results.** The decomposition (1) isolates a self-adjoint *reversible backbone*

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S, \tag{2}$$

which captures the reversible skeleton of the slow dynamics. Let $\lambda(Q_O)$ denote its spectral gap on the mean-zero slow subspace.

Within the DMS hypocoercivity framework [7, 6], we construct a modified Lyapunov functional $\mathcal{E}_{\alpha,\varpi}(f) = \frac{1}{2}\|f\|^2 + \alpha\, \Re\langle \mathcal{A}_\varpi f, f\rangle$ with an explicit corrector operator $\mathcal{A}_\varpi$ adapted from [11, 2], and establish the following.

Theorem A (Theorem 4.1). *Under Assumptions 3.1–3.3, for every $\gamma > 0$ and $\eta \le \eta^*(\gamma)$,*

$$\|e^{t\mathcal{L}} f\| \le \sqrt{3}\, e^{-\lambda_\eta t} \|f\|, \qquad \forall\, f \in \mathcal{H}_0,\ t \ge 0,$$

with an explicit rate $\lambda_\eta \ge \frac{2}{3}(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma))$. Here $\Lambda_{\text{hypo}}(\gamma)$ is the unperturbed hypocoercive rate, determined by the spectral gap $\lambda(Q_O)$, the fast dissipation gap λ_R, and the friction γ (see (38) *for the explicit formula), while $B_{\text{pert}}(\gamma)$ quantifies the destabilising effect of the non-equilibrium driving (see* (39)*). The stability threshold $\eta^*(\gamma) := \Lambda_{\text{hypo}}(\gamma)/(2B_{\text{pert}}(\gamma))$ is fully explicit and computable.*

Theorem B (Theorem 4.4). *Under the same assumptions, any admissible L^2 exponential rate ν with prefactor C satisfies the overdamped ceiling*

$$\nu \le C\left[\frac{(1+\kappa_{\text{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1{+}b_2) + \eta(a_1{+}a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

In particular, for $\gamma \to \infty$ and fixed η, the rate is at most $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$: the diffusive bottleneck persists, confirming that acceleration is inherently a finite-friction phenomenon.

Taken together, Theorems A and B give a clear picture. At intermediate friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, the rate scales as $\lambda_\eta \asymp \sqrt{\lambda(Q_O)}$, which is a quadratic speedup over the diffusive scale $\lambda(Q_O)$. This accelerated regime persists exactly while $\eta \le \eta^*(\gamma)$; inside that regime the rate deteriorates linearly in η. Theorem B shows that the large-friction ceiling remains $\mathcal{O}(\gamma^{-1})$, so once η exceeds threshold, moving to other friction regimes cannot recover kinetic scaling.

1.3. **Kinetic acceleration and non-equilibrium stability.** Our contribution has two parts. First, lifting gives a *quadratic speedup*: at optimal finite friction $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$, the rate improves from $\lambda(Q_O)$ to $\sqrt{\lambda(Q_O)}$. In the perturbative regime $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$ (Remark 3.4), this is substantial when the slow gap is small. Second, we give an explicit quantitative description of how non-equilibrium driving changes this acceleration; earlier lifting analyses [5, 9, 1, 18] focus on $\eta = 0$.

The rate degrades *linearly* in η, with slope $B_{\text{pert}}(\gamma)$. At optimal friction, Corollary 4.2 shows a key cancellation: the apparent $\lambda(Q_O)^{-1/2}$ singularity in B_{pert} is offset by the Lyapunov weight. As a result, $B_{\text{pert}}(\gamma_{\text{opt}}) = \mathcal{O}(1)$ and $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$. Interpretation: faster reversible mixing allows stronger non-equilibrium forcing while keeping kinetic acceleration.

An important subtlety is that the lifted NESS h_η and the overdamped NESS g_η may differ at finite friction: $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$ in general. However, for lifts constructed via the reverse adiabatic elimination (Section 5.1), one can always enforce $P_F \mathcal{V} P_F = 0$ as a free design choice, eliminating this bias entirely (Proposition 5.1). In this case $\varepsilon_* = 0$ (see Algorithm 1) and the full quadratic speedup applies for all target accuracies; a two-phase extension remains available for natural physical models where $P_F \mathcal{V} P_F \neq 0$ (Remark 5.3).

1.4. **Related work.** The hypocoercive method was pioneered by Villani [24] via modified H^1-type functionals. The L^2 approach based on modified Lyapunov functionals was developed by Dolbeault–Mouhot–Schmeiser [7, 6], who introduced the auxiliary corrector operator that restores coercivity on the degenerate subspace; this DMS framework is the methodological foundation of the present work.

The adaptation to Langevin dynamics in weighted Sobolev spaces, using resolvent-type correctors, was carried out by Iacobucci–Olla–Stoltz [11] and Cao–Lu–Wang [2].

The lifting perspective—embedding slow reversible dynamics into an enlarged state space with faster mixing—originates in the Markov chain literature with Chen–Lovász–Pak [5] and has been extended to continuous time by Eberle–Lörler [9] and Brigati–Lörler–Wang [1]. The quantum analogue was recently developed by Li–Lu [18]. All of these works assume the target (slow) dynamics is *reversible* (detailed balance). Our framework extends the theory to non-reversible targets via the perturbative decomposition $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ and the self-adjoint reversible backbone Q_O.

Quantitative Zeno-limit convergence in the singular regime $\gamma \to \infty$ was recently established by Carlen–Huse–Lebowitz [3] for boundary-driven quantum systems.

1.5. **Outline.** Section 2 sets up the abstract framework and adiabatic elimination. Section 3 formulates the structural assumptions and introduces the DMS corrector. Section 4 proves the main decay-rate bounds, including the stability threshold and singularity cancellation at optimal friction. Section 5 verifies the framework in two concrete models and develops the reverse adiabatic elimination recipe. Section 6 discusses extensions and open problems.

2. Preliminaries

We set up the abstract Hilbert-space framework and the adiabatic-elimination reduction used throughout.

2.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t\geq 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations. In the noncommutative (quantum) setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ on a finite-dimensional space $\mathfrak{h}$, equipped with the KMS inner product $\langle X, Y\rangle_\sigma := \operatorname{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$. The generator is then a GKSL Lindbladian $\mathcal{L}(X) = i[H, X] + \sum_\alpha \mathcal{D}^\dagger[L_\alpha]X$. In the classical (PDE) setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ.

2.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Decompose the generator as $\mathcal{L} = \mathcal{S} + \mathcal{A}$ with $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$ (symmetric part) and $\mathcal{A} := \frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ (skew part).

Definition 2.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f\in\mathcal{H}_0\setminus\{0\}}(-\langle f, \mathcal{S}f\rangle/\|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \operatorname{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\mathcal{A}$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [24, 7]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \leq Ce^{-\lambda t}\|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive.*

Lemma 2.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$*. Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive (*$C = 1$*). Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

2.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \leq 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = u + v$ with $u = P_S f$ (macroscopic) and $v = P_F f$ (microscopic).

Definition 2.3 (Centered coupling)**.** The interaction satisfies $P_S\mathcal{V}P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [19, 3].

Definition 2.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{3}$$

In the context of lifting, $\mathcal{L}_O$ is the effective "target" dynamics one seeks to accelerate. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau \mathcal{L}_O}$ as $\gamma \to \infty$ (quantitative Zeno-limit error bounds are given in [3]). From the rate perspective, Theorem 4.4 shows that the decay is bounded by a diffusive γ^{-1} ceiling in this overdamped regime, motivating the finite-friction analysis of Section 4.

When the interaction is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$ (skew-adjoint part plus perturbation), the centering condition is enforced componentwise: $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$.

3. The non-equilibrium lifting framework

We now state the structural assumptions used for quantitative hypocoercivity at finite friction. The goal is to isolate the ingredients that transfer dissipation from fast modes to slow modes through the DMS corrector [7, 6, 11]: fast spectral gap, centering, flux regularity, and perturbation bounds.

3.1. Structural assumptions.

Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \ge 0$:

$$\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}. \tag{4}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

The next assumption packages exactly the hypotheses needed in Section 4: fast coercivity, centered coupling, control of slow-induced fluxes, and energy-norm bounds for the perturbation.

Assumption 3.1 (Structural Conditions on the Non-Equilibrium Generator)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma \mathcal{R} + \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$. We assume the following structural conditions:

(1) **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R} f, f \rangle \ge \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{5}$$

In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \ge 0$ and $\|\mathcal{R}^+\| \le \lambda_R^{-1}$ (the bound follows from the spectral theorem: $\mathcal{R}|_{\mathcal{H}_F} \le -\lambda_R$ implies $\|(\mathcal{R}|_{\mathcal{H}_F})^{-1}\| \le \lambda_R^{-1}$, valid for both bounded and unbounded self-adjoint operators without requiring compact resolvent).

(2) **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{6}$$

Since η is a free parameter in our framework, we enforce this componentwise:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{7}$$

This matches the centered coupling condition (Definition 2.3) used to define the effective generator $\mathcal{L}_O$.

(3) **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reversible backbone defined in (16) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\mathrm{skew}} P_S \psi \in \mathcal{H}_F. \tag{8}$$

In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\mathrm{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on a common invariant core $\mathcal{C}$ for the operators involved (specified in the examples/appendices) and extended to the relevant energy domains by density. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \le M_R \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle, \qquad \forall \psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S. \tag{9}$$

Since (9) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \ge 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle (-\mathcal{R})h, h \rangle \le M_R \|h\|^2. \tag{10}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\mathrm{ell}} > 0$ such that for all $\psi \in \mathrm{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\mathrm{skew}} P_F \left(-\mathcal{R}^+\right) g_\psi\| \le \ \kappa_{\mathrm{ell}} \|P_S \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi\| \ = \ \kappa_{\mathrm{ell}} \|(-Q_O)\psi\|. \tag{11}$$

(4) **Structural Decomposition of the Perturbation.** The perturbation operator $\mathcal{V}_{\mathrm{pert}}$ admits a decomposition into a skew-symmetric part and a symmetric part, each controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}, \tag{12}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\mathrm{pert}}$ is skew-symmetric, i.e., for all $f \in \mathrm{dom}(\mathcal{V}_{\mathrm{pert}})$:

$$\Re\langle \mathcal{K}_{\mathrm{pert}} f, f \rangle = 0. \tag{13}$$

Moreover, $\mathcal{K}_{\mathrm{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \ge 0$ such that for all $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\mathrm{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{14}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\mathrm{pert}}$ is symmetric and defined at least on the energy domain $\mathrm{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \ge 0$ such that for all $f \in \mathrm{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\mathrm{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{15}$$

Equivalently, $\mathcal{S}_{\mathrm{pert}}$ is bounded as a map from $(\mathrm{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\mathrm{pert}}$ is often an unbounded multiplier on plain L^2; (15) is the standard replacement allowing it to be controlled by the fast dissipation.)

Under Assumption 3.1, we recall the effective generator from Definition 2.4, $\mathcal{L}_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion via the **reversible backbone** Q_O:

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S. \tag{16}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle \le 0. \tag{17}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

Remark 3.2 (Well-definedness of Q_O in unbounded settings). In finite dimensions the composition $Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$ is automatic, but in infinite-dimensional kinetic models $\mathcal{V}_{\text{skew}}$ is an unbounded differential operator and one must check that $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi \in \text{dom}(\mathcal{V}_{\text{skew}})$. This is ensured by Assumption 3.1.3, which stipulates that all compositions are interpreted on a common invariant core $\mathcal{C}$ (e.g. Schwartz functions) preserved by all operators involved; in the Langevin example, the pseudoinverse $\mathcal{R}^+$ is smoothing in the momentum variable, so the intermediate output $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi$ falls in the domain of the second application of $\mathcal{V}_{\text{skew}}$, and Q_O extends to a well-defined self-adjoint operator by integration by parts on $\mathcal{C}$.

Assumption 3.3 (Spectral Gap of Effective Dynamics)**.** Let $\mathcal{K} := \ker(Q_O) \subset \mathcal{H}_S$ be the kernel of the reference generator. We assume Q_O admits a **spectral gap** $\lambda(Q_O) > 0$ on the orthogonal complement $\mathcal{K}^\perp := \mathcal{H}_S \ominus \mathcal{K}$. That is, the Poincaré inequality holds:

$$-\langle f, Q_O f\rangle \ge \lambda(Q_O)\|f - P_{\mathcal{K}} f\|^2, \quad \forall f \in \mathcal{H}_S. \tag{18}$$

Consequently, for any shift parameter $\varpi > 0$, the operator $(\varpi - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\text{dom}(Q_O)$.

Remark 3.4 (Gap equivalence: $\mathcal{L}_O$ vs. Q_O). By the expansion (57),

$$\mathcal{L}_O = Q_O + \eta\, \mathcal{B}_1 + \eta^2\, \mathcal{B}_2, \tag{19}$$

with model-dependent bounded operators $\mathcal{B}_1, \mathcal{B}_2$ on $\mathcal{H}_S$ in the finite-dimensional setting (and relatively bounded perturbations on the common core in the unbounded examples). Hence

$$\|\mathcal{L}_O - Q_O\| \le c_1\eta + c_2\eta^2 \tag{20}$$

for suitable constants $c_1, c_2 > 0$. If the gap $\lambda(Q_O)$ is isolated and η is such that

$$c_1\eta + c_2\eta^2 \le \theta\, \lambda(Q_O), \qquad 0 < \theta < 1, \tag{21}$$

then resolvent perturbation theory for isolated spectral sets [13] gives persistence of the stable gap:

$$(1-\theta)\lambda(Q_O) \le \lambda(\mathcal{L}_O) \le (1+\theta)\lambda(Q_O). \tag{22}$$

(Here $\lambda(\mathcal{L}_O)$ denotes the gap measured on the stable part when $\mathcal{L}_O$ is non-self-adjoint.) In particular, in the perturbative regime relevant to this paper, $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$, so the convergence estimate $\lambda_\eta \asymp \sqrt{\lambda(Q_O)} \approx \sqrt{\lambda(\mathcal{L}_O)}$ is consistent.

Remark 3.5 (Scope: self-adjointness excludes Liouvillian Exceptional Points). Assumption 3.3 postulates $\lambda(Q_O) > 0$, which in particular requires Q_O to have a complete set of eigenfunctions (no defective spectrum). Liouvillian Exceptional Points (LEPs)—where two eigenvalues coalesce and the generator becomes non-diagonalizable—are therefore excluded by hypothesis. In our Hilbert-space setting this is not an additional restriction: the self-adjointness of Q_O established above already guarantees diagonalizability by the spectral theorem, so genuine LEPs can only arise in the non-self-adjoint target effective generator $\mathcal{L}_O$ [21], not in Q_O.

3.2. **The DMS corrector.** To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. Following the Dolbeault–Mouhot–Schmeiser (DMS) approach [7, 6], extended to L^2 settings by Iacobucci–Olla–Stoltz [11] and Cao–Lu–Wang [2], this operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level.

Definition 3.6 (DMS Corrector)**.** Let $\varpi > 0$ be a shift parameter, to be optimized in Section 4; it is distinct from the reference measures μ_0, μ_η appearing in the examples. We define the **corrector** $\mathcal{A}_\varpi : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F. \tag{23}$$

The key obstruction to a direct energy estimate is that dissipation is degenerate on $\mathcal{H}_S$: the plain energy $\frac{1}{2}\|f\|^2$ cannot produce a spectral gap when $\ker(\mathcal{R})$ is non-trivial.

Proposition 3.7 (Why the plain energy method fails)**.** *Assume $\eta = 0$ and $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}}$ with $\mathcal{V}_{\text{skew}}$ skew-adjoint and $\ker(\mathcal{R}) = \mathcal{H}_S \neq \{0\}$. Then there is no constant $\lambda > 0$ such that*

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2 \leq -\lambda\,\|f_t\|^2 \qquad \textit{for all solutions } \partial_t f_t = \mathcal{L} f_t. \tag{24}$$

Proof. For any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\mathcal{R}\phi = 0$ and $\Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0$, hence

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2\Big|_{t=0,\, f_0=\phi} = \Re\langle \mathcal{L}\phi, \phi\rangle = \gamma\langle \mathcal{R}\phi, \phi\rangle + \Re\langle \mathcal{V}_{\text{skew}}\phi, \phi\rangle = 0. \tag{25}$$

If such a $\lambda > 0$ existed, we would get $0 \leq -\lambda\|\phi\|^2 < 0$, a contradiction. □

Proposition 3.7 confirms that the energy $\frac{1}{2}\|f\|^2$ alone is degenerate on $\mathcal{H}_S$. The remedy is to augment it with a cross term involving the corrector $\mathcal{A}_\varpi$; differentiating $\Re\langle \mathcal{A}_\varpi f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\varpi$, including the control over non-equilibrium perturbations.

Lemma 3.8 (Structural Estimates for the DMS Corrector)**.** *Let $\varpi > 0$ be the resolvent shift parameter. The corrector $\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ satisfies the following structural estimates, where Q_O is the reference effective dynamics.*

Under Assumption 3.1, the operator $\mathcal{A}_\varpi$ satisfies the following structural estimates uniformly in ϖ. The constants defined below depend solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$:

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \leq k_{\mathcal{A}}\varpi^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{26}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi\| \leq k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{27a}$$

$$\|\mathcal{A}_\varpi\mathcal{V}_{\text{skew}}P_F\| \leq k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{27b}$$

The dual estimate (27b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 3.1.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\varpi\| \leq k_P(\varpi), \tag{28a}$$

$$\|\mathcal{A}_\varpi\mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\varpi). \tag{28b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}}\, \varpi^{-1/2}, \tag{29}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2}. \tag{30}$$

Moreover, the centered perturbation coupling in Assumption 3.1.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^{\dagger} P_S = 0, \tag{31}$$

and therefore

$$\mathcal{V}_{\text{pert}}\, \mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^{\dagger}\, \mathcal{H}_S \subset \mathcal{H}_F. \tag{32}$$

Proof. See Appendix B. □

3.3. **The modified Lyapunov functional.** We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the corrector $\mathcal{A}_\varpi$.

Definition 3.9 (Modified Lyapunov functional [7, 6])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha \Re\langle \mathcal{A}_\varpi f, f\rangle. \tag{33}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\varpi$.

Proposition 3.10 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{34}$$

Proof. The proof follows directly from the Cauchy-Schwarz inequality $|\Re\langle \mathcal{A}_\varpi f, f\rangle| \le \|\mathcal{A}_\varpi\|\|f\|^2$ and the triangle inequality. □

4. Convergence bounds for the lifted semigroup

We now derive quantitative decay bounds for the semigroup on the mean-zero subspace. The strategy has two parts: a constructive lower bound from a 2×2 dissipation matrix, and a complementary overdamped upper ceiling from singular values. Together they identify the finite-friction acceleration window.

4.1. **Lower bound via modified hypocoercivity.** This subsection gives an explicit lower bound on the decay rate by jointly optimizing the Lyapunov parameters (α, ϖ). The optimization is fully algebraic and returns both the friction-dependent profile and the admissible perturbation range.

The complete matrix estimates and parameter optimization are deferred to Appendix C; the explicit formulas used throughout the sequel are stated below.

Notation for the rate functions. The following structural scalars, depending only on the parameters in Assumption 3.1, are fixed throughout this section:

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{35}$$

where $k_{\text{reg}} = \sqrt{M_R/\lambda_R}$ (see (27a)) and κ_{ell} is from (11). Interpretation: Θ measures slow–fast flux amplitude, K_{geom} collects coupling geometry constants, and $b_{2,\text{eff}}$ records the fast-mode amplification from $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R \|f_F\|^2$.

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{36}$$

which measures effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{37}$$

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}, \tag{38}$$

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}, \tag{39}$$

where $C_{\text{sing}} := 2(b_1 + b_2)\, k_{\mathcal{A}} + \frac{3}{4}(a_1 + a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 3.8. Here $\alpha_*(\gamma)$ is the optimal Lyapunov mixing weight, $\Lambda_{\text{hypo}}(\gamma)$ is the lower-rate profile, and $B_{\text{pert}}(\gamma)$ quantifies perturbation destabilization. Although B_{pert} contains an apparent $\lambda(Q_O)^{-1/2}$ factor, it is cancelled at optimal friction because $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ (Corollary 4.2).
Main result. The structural properties established in Appendix C yield a uniform norm equivalence (Proposition 3.10) with prefactor $C = \sqrt{3}$ for all $\gamma > 0$.

Theorem 4.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 3.1 and 3.3 hold. Set $\eta^*(\gamma) := \Lambda_{\text{hypo}}(\gamma)/(2B_{\text{pert}}(\gamma))$. For any $\gamma > 0$, if $\eta \le \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on $\mathcal{H}_0$:*

$$\|e^{t\mathcal{L}} f\| \le \sqrt{3}\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{40}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{2}{3}\left(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\right). \tag{41}$$

Sketch of Proof. The full computation is in Appendix C. With the resolvent shift fixed at $\varpi_* = \lambda(Q_O)$, the Lyapunov derivative satisfies $\frac{d}{dt}\mathcal{E}_{\alpha,\varpi_*} \le -\mathbf{v}^T\mathbf{M}(\alpha)\mathbf{v}$ for the vector $\mathbf{v} = (\|P_F f\|, \|P_S f\|)^T$ and a 2×2 positive-definite dissipation matrix $\mathbf{M}(\alpha)$. Maximising over α yields the optimal weight $\alpha_*(\gamma)$ from (37) and the rate $\Lambda_{\text{hypo}}(\gamma)$. The distortion parameter $\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|$ satisfies

$$\delta(\gamma) \le \frac{\sqrt{\lambda_R}}{4\Theta K_{\text{geom}}} \le \tfrac{1}{2} \qquad \forall\, \gamma > 0, \tag{42}$$

by the solver bound $\|\mathcal{A}_{\varpi_*}\| \le (2\sqrt{\lambda_R})^{-1}\lambda(Q_O)^{-1/2}$, the formula for α_* in (37), and the elementary extremum $\sup_{x>0} x(x + K_{\text{geom}})^{-2} = (4K_{\text{geom}})^{-1}$. This gives the uniform equivalence $\frac{1}{4}\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*} \le \frac{3}{4}\|f\|^2$, hence the prefactor $\sqrt{3}$ and the $\frac{2}{3}$ factor in (41). □

Regime summary. The formula (38) has two clear regimes. For $\gamma \to \infty$, the denominator behaves like $Z(\gamma)^2 \sim \gamma^2\Theta^2\lambda(Q_O)^{-1}$, so $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$ (diffusive). For small or order-one γ, the denominator is dominated by the constant part $K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}$, so $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$. Hence the maximum is attained at intermediate friction and scales as $\max_\gamma \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

Corollary 4.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 4.1. Assume furthermore that the structural constants Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\text{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O) \downarrow 0$), and that $b_{2,\text{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\text{hypo}}(\gamma)$ defined in (38) attains its global maximum at

$$\gamma_{\text{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\text{geom}}^2 + \left(2k_{\text{reg}} + \tfrac{1}{2}\right)} \asymp \sqrt{\lambda(Q_O)}. \tag{43}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{44}$$

Moreover, the mixing weight in (39) *satisfies*

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{45}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\mathrm{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and the stability threshold satisfies

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{46}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\left(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\right)^2 + \left(2k_{\mathrm{reg}} + \frac{1}{2}\right)}. \tag{47}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\mathrm{reg}} + \frac{1}{2}$. Then $\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\mathrm{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\mathrm{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (39), $B_{\mathrm{pert}}(\gamma) = b_{2,\mathrm{eff}} + \alpha_*(\gamma)\,C_{\mathrm{sing}}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\mathrm{opt}}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\mathrm{opt}}) = x + K_{\mathrm{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\mathrm{opt}}) = \frac{\gamma_{\mathrm{opt}}\lambda_R}{Z(\gamma_{\mathrm{opt}})^2+2k_{\mathrm{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$. Finally, $\eta^*(\gamma) = \Lambda_{\mathrm{hypo}}(\gamma)/(2B_{\mathrm{pert}}(\gamma))$ yields $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$. □

Define the (rigorous) L^2 relaxation time as in [18] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \ge 0\,;\ \|e^{t\mathcal{L}}f\| \le e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{48}$$

The following holds:

Corollary 4.3 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem 4.1. Then for $\eta \le \eta^*(\gamma)$, the exponential decay estimate from Theorem 4.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \le \frac{1+\log\sqrt{3}}{\lambda_\eta}. \tag{49}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ (Corollary 4.2) and for perturbations within the stability threshold, one has

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{50}$$

Proof. By Theorem 4.1, for all $f \in \mathcal{H}_0$ and $t \ge 0$ one has $\|e^{t\mathcal{L}}f\| \le \sqrt{3}\,e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1+\log\sqrt{3})/\lambda_\eta$ so that $\sqrt{3}\,e^{-\lambda_\eta t} = e^{-1}$. □

4.2. **Overdamped diffusive ceiling.** To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f\in\mathcal{H}_0\cap\mathrm{dom}(\mathcal{L})\setminus\{0\}} \frac{\|\mathcal{L}f\|}{\|f\|}, \tag{51}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversiblizations* and the *pseudo spectral gap* used in non-reversible Markov

chain theory; see, e.g., [20]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$ [4].

Theorem 4.4 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumptions 3.1.1–4 and Assumption 3.3. Assume also* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume moreover that* $\lambda(Q_O)$ *is attained by a normalized eigenfunction* $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ *(automatic in finite dimensions, and in many PDE settings with compact resolvent). Let* $C \geq 1$ *and* $\nu > 0$ *be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{52}$$

Then for every $\gamma > 0$*,*

$$\nu \leq C\, s(\mathcal{L}) \leq C \left[\frac{(1+\kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{53}$$

where κ_{ell} *is the flux-regularity constant from Assumption 3.1.3 and* a_1, a_2, b_1, b_2 *are from Assumption 3.1.4. In particular, for* $\eta \leq c/\gamma$ *one obtains the diffusive overdamped scaling* $\nu = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$*.*

Sketch of Proof. The proof is given in Appendix D. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$. The flux regularity bound (11) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\mathrm{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption 3.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. □

Theorem 4.4 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem 4.1, this confirms that the accelerated $\sqrt{\lambda(Q_O)}$ regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

Corollary 4.5 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem 4.4. Define* $t_{\mathrm{rel}}(\mathcal{L})$ *as in Corollary 4.3. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \leq \frac{(1+\kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{54}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2, with the same prefactor C *as in Theorem 4.4):*

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2C\, s(\mathcal{L})}. \tag{55}$$

Consequently, if $\eta \leq c/\gamma$*, then* $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ *as* $\gamma \to \infty$ *and hence*

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{56}$$

In particular, the kinetic scaling $t_{\mathrm{rel}}(\mathcal{L}) \asymp \lambda(Q_O)^{-1/2}$ *is achieved at optimal friction, combined with Corollary 4.3.*

5. Examples

This section has two goals. First, we give a direct construction that starts from a target slow generator $\mathcal{L}_O$ and builds a lifted generator $\mathcal{L}$ whose adiabatic elimination returns $\mathcal{L}_O$. Second, we apply this construction to two examples: forced underdamped Langevin dynamics (Section 5.2) and a finite-dimensional boundary-driven model (Section 5.3).

5.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** Input: a non-equilibrium effective generator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$. Output: a lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ such that adiabatic elimination (Definition 2.4) gives back $\mathcal{L}_O$. We call this construction *reverse adiabatic elimination*. The procedure has three steps.
Perturbation expansion of $\mathcal{L}_O$. By Definition 2.4, $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$. Expanding in powers of η gives

(57)
$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\,P_S}_{Q_O}\ \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\ +\ \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S}_{\text{cross terms }\mathcal{O}(\eta)}\ \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S}_{\text{pure perturbation }\mathcal{O}(\eta^2)}.$$

The first term is the reversible backbone Q_O (the $\eta = 0$ limit). The second and third terms are the $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ corrections. In the boundary-driven example, all these terms are written explicitly in (74).
Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The resulting operator is Q_O. Its spectral gap $\lambda(Q_O)$ is the key scale in Theorem 4.1 (both for accelerated rate and for the admissible perturbation size).
Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = -T^*\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\text{skew}}P_S, \tag{58}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^*\mathcal{R}_F^{-1}T = -Q_O$. A universal choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\text{skew}} := \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$.
Step 3: Determine $\mathcal{V}_{\text{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\text{skew}}$ fixed, we choose $\mathcal{V}_{\text{pert}}$ so that adiabatic elimination exactly reproduces $\mathcal{L}_O$. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O\ =\ -\eta\,P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S\ -\ \eta^2\,P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S, \tag{59}$$

with the centering constraint $P_S\mathcal{V}_{\text{pert}}P_S = 0$. In the Langevin example, $\mathcal{V}_{\text{pert}}P_S = 0$, so the formula simplifies strongly. In the boundary-driven example, all terms are present, and we write them explicitly in (74). The resulting lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ automatically satisfies centering and is suitable for Theorem 4.1. We now state the key structural advantage: for a broad class of designed lifts, the projection bias can be removed exactly.

Proposition 5.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (59)*. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the Schur complement (61) *satisfies $\gamma\,S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*
(iii) the lifted NESS satisfies $P_Sh_\eta = g_\eta$ for all $\gamma > 0$ and $\eta \le \eta^*(\gamma)$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F\mathcal{V}_{\text{pert}}P_F = 0$ is a free design choice: it does not appear in the consistency condition (59) *(which constrains only $P_F\mathcal{V}_{\text{pert}}P_S$ and $P_S\mathcal{V}_{\text{pert}}P_F$), so it can always be enforced without affecting the effective dynamics.*

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F\mathcal{V}_{\text{skew}}P_F = 0$. Combined with $P_F\mathcal{V}_{\text{pert}}P_F = 0$, we get $P_F\mathcal{V}P_F = P_F(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_F = 0$. (ii) With $P_F\mathcal{V}P_F = 0$, the Schur complement (61) reduces to $S(\gamma) = -P_S\mathcal{V}(\gamma\mathcal{R}_F)^{-1}\mathcal{V}\,P_S = \gamma^{-1}\mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma)u = 0$ becomes $\mathcal{L}_Ou = 0$, so $u = g_\eta$. □

Proposition 5.1 gives a clean design rule: if we enforce $P_F\mathcal{V}_{\text{pert}}P_F = 0$, then the projected lifted steady state equals the overdamped steady state for all γ. This does *not* remove non-equilibrium

effects, because $\mathcal{L}_O$ still depends on the off-diagonal perturbation blocks $P_S\mathcal{V}_{\text{pert}}P_F$ and $P_F\mathcal{V}_{\text{pert}}P_S$. So non-self-adjoint slow dynamics remain fully allowed.

Remark 5.2 (Kernel consistency and invariant states). We verify that the lifted and effective dynamics share the same stationary state.

Equilibrium case ($\eta = 0$): exact identity for all $\gamma > 0$. Let $f \in \ker(\mathcal{L}|_{\eta=0}) = \ker(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}})$. Taking $\operatorname{Re}\langle f, \mathcal{L}f\rangle = \gamma\langle f, \mathcal{R}f\rangle = 0$ (using skew-adjointness of $\mathcal{V}_{\text{skew}}$) forces $f \in \ker(\mathcal{R}) = \mathcal{H}_S$. Then $\mathcal{V}_{\text{skew}}f = 0$, which entails $Tf = P_F\mathcal{V}_{\text{skew}}P_Sf = 0$ and thus $Q_Of = -T^*\mathcal{R}_F^{-1}Tf = 0$. Therefore

$$\ker(\mathcal{L}|_{\eta=0}) = \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \forall\, \gamma > 0. \tag{60}$$

This is the analogue of [18, Lemma 2.10, eq. (2.17)].

Non-equilibrium case ($\eta \neq 0$). When $\eta \neq 0$ and $P_F\mathcal{V}P_F \neq 0$, the NESS $h_\eta \in \ker(\mathcal{L})$ acquires a fast component $P_Fh_\eta = \mathcal{O}(\eta/\gamma)$, and the slow marginal $u = P_Sh_\eta$ satisfies the Schur complement equation

$$S(\gamma)\, u = 0, \qquad S(\gamma) := -P_S\mathcal{V}\big(\gamma\mathcal{R}_F + P_F\mathcal{V}P_F\big)^{-1}\mathcal{V}\, P_S. \tag{61}$$

Expanding in a Neumann series:

$$\gamma\, S(\gamma) = \mathcal{L}_O + \gamma^{-1}\, P_S\mathcal{V}\,\mathcal{R}^+\,(P_F\mathcal{V}P_F)\,\mathcal{R}^+\,\mathcal{V}\,P_S + \mathcal{O}(\gamma^{-2}). \tag{62}$$

Thus $P_Sh_\eta \to g_\eta$ as $\gamma \to \infty$ (the Zeno/adiabatic limit). The Schur correction is of order $\mathcal{O}(\gamma^{-1})$. Under the standard perturbative setting (simple isolated zero mode and smooth dependence on η), this gives $\Delta_{\text{bias}}(\gamma) := \|P_Sh_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. However, when $P_F\mathcal{V}P_F = 0$ (as guaranteed by Proposition 5.1 for designed lifts), the expansion collapses to $\gamma\, S(\gamma) = \mathcal{L}_O$ and $\Delta_{\text{bias}}(\gamma) \equiv 0$.

Since the lifted dynamics converges to h_η, the approximation error for the overdamped NESS g_η satisfies the triangle inequality

$$\|P_Se^{t\mathcal{L}}f - g_\eta\|_{\mathcal{H}_S} \;\le\; \underbrace{e^{-\lambda_\eta t}\,\|f - h_\eta\|}_{\text{convergence error}} \;+\; \underbrace{\|P_Sh_\eta - g_\eta\|_{\mathcal{H}_S}}_{\text{bias } \Delta_{\text{bias}}(\gamma)}. \tag{63}$$

For lifts satisfying the conditions of Proposition 5.1, $\Delta_{\text{bias}}(\gamma) = 0$ and the accelerated mixing time is

$$T_{\text{mix}}(\varepsilon) \;=\; \mathcal{O}\Big(\frac{\log(1/\varepsilon)}{\sqrt{\lambda(Q_O)}}\Big), \qquad \forall\, \varepsilon > 0. \tag{64}$$

Algorithm 1 summarises the procedure.

Remark 5.3 (Two-phase extension for general models). For natural physical models where $P_F\mathcal{V}P_F \neq 0$ (and the conditions of Proposition 5.1 do not hold), a nonzero accuracy floor $\varepsilon_* := \Delta_{\text{bias}}(\gamma_{\text{opt}}) = \mathcal{O}(\eta/\sqrt{\lambda(Q_O)})$ may arise. Then the error bound (63) cannot go below ε_* using only lifted dynamics. A direct remedy is two-phase evolution: first run the lifted dynamics at γ_{opt} down to scale ε_*, then switch to $\mathcal{L}_O$ to reach the final target ε. The total time is at most the overdamped baseline and is strictly smaller whenever $\lambda_\eta > \lambda(\mathcal{L}_O)$. This extension is not needed for either of the examples in this paper (Remark 5.6).

Remark 5.4 (Connection to the literature). Our construction has the same lifting spirit as previous acceleration methods in Markov-chain and MCMC settings; see [22, 17, 10, 18]. The present formulation is fully continuous-time and expressed in the operator framework of Assumption 3.1.

Algorithm 1 Accelerated NESS Approximation via Lifting

Require: Target effective generator $\mathcal{L}_O$ on $\mathcal{H}_S$, target accuracy $\varepsilon > 0$
Ensure: Approximation of g_η (overdamped NESS) to accuracy ε

Step 1: Construct lifted dynamics (Section 5.1)
Factor $Q_O = -T^*\mathcal{R}_F^{-1}T$ and set $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$
Choose $\mathcal{V}_{\text{pert}}$ satisfying (59) with $P_F\mathcal{V}_{\text{pert}}P_F = 0$ ▷ Zero bias (Prop. 5.1)

Step 2: Compute optimal friction
$\gamma_{\text{opt}} \leftarrow \sqrt{\lambda(Q_O)} \cdot \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}$
$\lambda_\eta \leftarrow \frac{2}{3}\big(\Lambda_{\text{hypo}}(\gamma_{\text{opt}}) - \eta\, B_{\text{pert}}(\gamma_{\text{opt}})\big)$

Step 3: Run lifted dynamics (rate $\sim \sqrt{\lambda(Q_O)}$)
$T \leftarrow \log\big(\sqrt{3}\|f_0\|/\varepsilon\big)/\lambda_\eta$
return $P_S\big(e^{T\mathcal{L}}h_0\big)$ ▷ ε-approximation of g_η

5.2. **Example 1: Classical Nonequilibrium Langevin Dynamics.** We now instantiate the framework on forced underdamped Langevin dynamics on $\mathbb{T}^d \times \mathbb{R}^d$.

Setup. We consider a particle with conservative potential $U \in C^\infty(\mathbb{T}^d)$ and non-conservative force $F \in C^\infty(\mathbb{T}^d; \mathbb{R}^d)$. Let μ_0 be the Gibbs measure of the unperturbed dynamics:

$$d\mu_0(q,p) = Z^{-1}e^{-\beta(\frac{1}{2}|p|^2 + U(q))}dq\,dp, \tag{65}$$

with factorization $d\mu_0 = d\nu(q)\,d\nu_p(p)$. We work in $\mathcal{H} = L^2(\mu_0)$. For $\eta \neq 0$, convergence to the NESS is measured in $L^2(\mu_0)$ toward the density h_η. All identities are first checked on the core $\mathcal{C}_{\text{Lan}} := C^\infty(\mathbb{T}^d) \otimes \mathcal{S}(\mathbb{R}^d)$ and then extended by density.

Generator decomposition. With slow projection $P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\,d\nu_p(p)$, the generator decomposes as $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$:

$$\mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla U(q) \cdot \nabla_p, \qquad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f. \tag{66}$$

Effective slow generator. Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\,d\nu_p = 0$, one computes via Definition 2.4:

$$\mathcal{L}_O\phi = \beta^{-1}\Delta_q\phi - (\nabla U - \eta F) \cdot \nabla_q\phi, \qquad \phi = \phi(q), \tag{67}$$

and the reference operator is $Q_O := \mathcal{L}_O|_{\eta=0} = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q$, self-adjoint on $L^2(\nu)$ with Dirichlet form $\langle \phi, (-Q_O)\phi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q\phi\|^2_{L^2(\nu)}$.

Proposition 5.5 (Verification of Assumption 3.1)**.** *The decomposition* (66) *satisfies Assumption 3.1.1–4 in $L^2(\mu_0)$, with structural constants:*

(i) $\lambda_R = 1$, $M_R = 1$ *(Ornstein–Uhlenbeck spectral gap and conditioning);*
(ii) $\kappa_{\text{ell}} = \sqrt{d(d+1)}\,\sqrt{\rho_{\max}/\rho_{\min}} \cdot (1 + \beta\|\nabla U\|_{L^\infty}\sqrt{C_{\text{P}}(\nu)})$, *where* $\rho_{\max}/\rho_{\min} = e^{\beta(\max U - \min U)}$ *and* $C_{\text{P}}(\nu)$ *is the Poincaré constant of* ν*;*
(iii) the perturbation constants (a_1, b_1, a_2, b_2) *depend only on* $(\beta, d, \|F\|_{L^\infty})$*, independently of* $\lambda(Q_O)$*.*

The proof (verification of skew-adjointness, centering, flux regularity via Fourier–Parseval analysis, and p-multiplier energy bounds) is given in Appendix E.

Scaling consequences. Because the structural constants do not depend on $\lambda(Q_O)$, Corollary 4.2 applies directly. Hence $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$, $\eta^*(\gamma_{\text{opt}}) \asymp C(\beta, d, \|F\|_{L^\infty})\sqrt{\lambda(Q_O)}$, and $\max_\gamma \Lambda_{\text{hypo}}(\gamma) \asymp$

$\sqrt{\lambda(Q_O)}$. This is consistent with the classical underdamped/overdamped crossover in [11], and adds an explicit admissible range for non-equilibrium forcing.

Numerical verification. We validate these predictions with Galerkin spectral computations in the Fourier–Hermite basis [11]. To keep orthonormality in $L^2(\mu_0)$, we use Boltzmann-weighted Fourier modes:

$$G_k(q) = \sqrt{\frac{Z_\nu}{2\pi}}\, e^{ikq+\beta U(q)/2}, \qquad Z_\nu = \int_0^{2\pi} e^{-\beta U}\, dq,$$

so that $\{\psi_{nk}(q,p) := G_k(q)H_n(p)\}$ is orthonormal in $L^2(\mu_0)$. For the rotor potential $U(q) = U_0(1-\cos q)$, the matrix elements are (with $m=1$):

(68)

$$\langle \psi_{n'k'}, \mathcal{L}\,\psi_{nk}\rangle = \sqrt{\beta(n+1)}\,\delta_{n',n+1}\Big[\tfrac{ik}{\beta}\delta_{kk'} + \tfrac{u_{k-k'}}{2}\Big] + \sqrt{\beta n}\,\delta_{n',n-1}\Big[\big(\tfrac{ik}{\beta}+\eta F\big)\delta_{kk'} - \tfrac{u_{k-k'}}{2}\Big] - \gamma n\,\delta_{nn'}\delta_{kk'},$$

where u_j are Fourier coefficients of $U'(q)$ (for the rotor, $u_{\pm 1} = \mp iU_0/2$). The numerical setup is: $d=1$, $\beta=100$, $F=0.05$, $K=8$, $N=16$, $U_0 \in \{0.01, 0.02, 0.05, 0.08, 0.1\}$, $\gamma \in [0.05, 3.0]$ (26 points), $\eta \in \{0, 0.1, 1\}$.

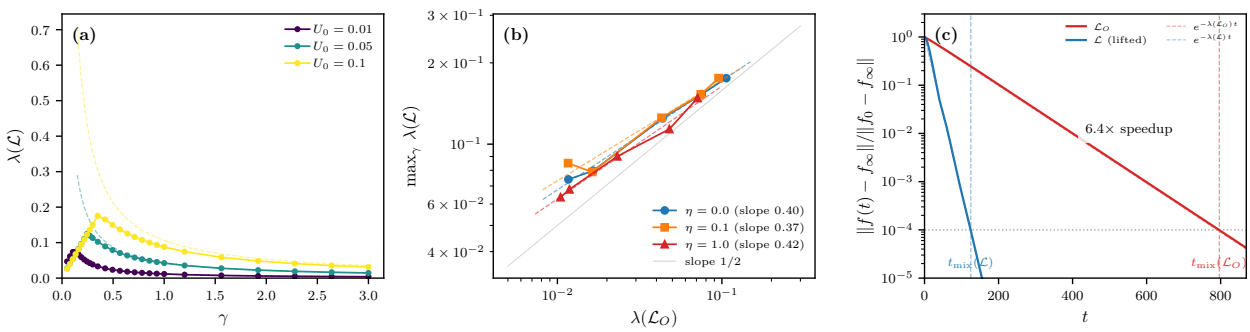

FIGURE 1. Numerical verification for Example 5.2. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of U_0; dashed curves show the overdamped ceiling $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$ for $\eta \in \{0, 0.1, 1\}$; fitted slopes are shown in the legend. **(c)** Semigroup decay for the lifted dynamics at γ_{opt} and for $\mathcal{L}_O$ ($U_0 = 0.01$, $\eta = 0$); dashed lines show the exponential envelopes $e^{-\lambda(\mathcal{L})\,t}$ and $e^{-\lambda(\mathcal{L}_O)\,t}$. Vertical markers indicate the ε-mixing times ($\varepsilon = 10^{-4}$). Parameters: $\beta = 100$, $F = 0.05$, $K = 8$, $N = 16$.

As a discretization check, the free case $U_0 = 0$ reproduces $\lambda_{n,k} = n\gamma + k^2/(\beta\gamma)$ [15] with relative error $\lesssim 10^{-15}$. We also compare with an unweighted Fourier–Hermite basis. Agreement is excellent for moderate βU_0, while at $\beta U_0 = 10$ the unweighted basis develops spurious unstable modes. This confirms that the weighted basis in (68) is the correct stable choice in our parameter range.

Figure 1(a) shows the expected three regimes: small-γ growth, large-γ diffusive ceiling, and an intermediate optimum. Figure 1(b) shows $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$ with fitted slope near 1/2, consistent with $\max_\gamma \lambda(\mathcal{L}) \propto \sqrt{\lambda(\mathcal{L}_O)}$. Figure 1(c) gives a representative mixing-time comparison at $\varepsilon = 10^{-4}$: for $U_0 = 0.01$, $\eta = 0$, we obtain $t_{\text{mix}}(\mathcal{L}) \approx 125$ versus $t_{\text{mix}}(\mathcal{L}_O) \approx 796$. No numerical accuracy floor is observed, and at $\eta = 0.1$ we find $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$.

Algorithmic lift via generic square-root factorization. For the reviewer, the construction used in this example is the following three-step procedure.

Step 1 (compute slow operators). From the Galerkin matrix in (68), we compute the slow generator $\mathcal{L}_O$ and its reversible part Q_O.

Step 2 (build a generic lifted fast block). We choose $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\lambda_R \,\mathrm{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This gives a 34×34 lifted generator (compared with 272×272 for the physics-adapted lift).

Step 3 (match the non-equilibrium part). We determine the remaining perturbation block by enforcing the consistency relation (59). In Langevin, $\mathcal{V}_{\mathrm{pert}} P_S = 0$ (because $F \cdot \nabla_p$ vanishes on the $n = 0$ Hermite mode), so this step reduces to a linear solve, which we compute directly.

Figure 2 compares the algorithmic and physical lifts. Unlike the boundary-driven model (Section 5.3), where both constructions coincide, here the two lifts differ: the physics-adapted lift uses the full momentum space $\mathcal{H}_F = L^2(\nu_p)$, whereas the algorithmic lift compresses it to a single copy of the slow space. To align the protocol with the boundary-driven algorithmic-lift experiment, we benchmark the same number of *non-equilibrium* configurations: $U_0 \in \{0.01, 0.02, 0.05, 0.08\}$ and $\eta \in \{0.10, 0.15\}$ (8 combinations in total). Across this matched grid, both lifts remain faster than the overdamped generator in peak-gap terms, and the log–log trends in Panel (b) remain consistent with $\sqrt{\lambda(\mathcal{L}_O)}$ scaling. Denoting the NESS of the algorithmic lift by h_η^{alg}, we verify $\|P_S h_\eta^{\mathrm{alg}} - g_\eta\|/\|g_\eta\| \lesssim 10^{-15}$ at $\eta = 0.1$.

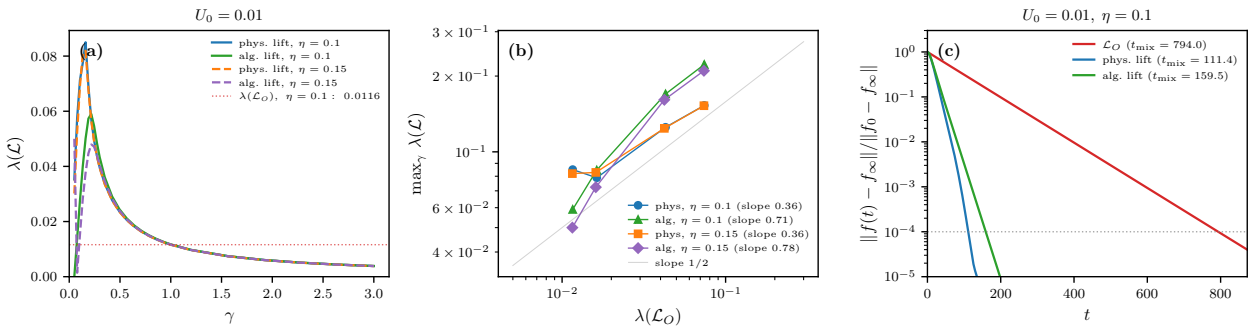

FIGURE 2. Comparison of the *physical* (272×272, full Langevin) and *algorithmic* (34×34, square-root factorization) lifted generators for Example 5.2. **(a)** Spectral gap vs. γ at $U_0 = 0.01$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped gap $\lambda(\mathcal{L}_O)$ is shown as a horizontal reference. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$; fitted slopes are listed in the legend. **(c)** Mixing-time comparison ($U_0 = 0.01$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 1. Peak spectral gaps for the physical and algorithmic lifts in the Langevin model on the matched non-equilibrium grid $U_0 \in \{0.01, 0.02, 0.05, 0.08\}$, $\eta \in \{0.10, 0.15\}$. The "speedup" column is $\max_\gamma \lambda(\mathcal{L})/\lambda(\mathcal{L}_O)$.

U_0	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	speedup
0.01	0.10	0.01161	0.08502	0.05904	5.09×
0.01	0.15	0.01156	0.08196	0.05000	4.33×
0.02	0.10	0.01629	0.07888	0.08460	5.19×
0.02	0.15	0.01608	0.08284	0.07238	4.50×
0.05	0.10	0.04292	0.12520	0.16978	3.96×
0.05	0.15	0.04220	0.12397	0.16097	3.81×
0.08	0.10	0.07405	0.15291	0.22225	3.00×
0.08	0.15	0.07369	0.15250	0.20992	2.85×

5.3. **Example 2: A Finite-Dimensional Boundary-Driven Toy Model.** We now test the framework on a two-qubit boundary-driven model. Its main role is to show that the method still works when the effective slow generator is non-self-adjoint. Because $\dim \mathcal{H} = 16$, all operators are bounded and there are no domain subtleties.

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\mathrm{pert}}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, so $\lambda_R = 4\Gamma$. The transport part is $\mathcal{V}_{\mathrm{skew}} X = -i[g\sigma_z \otimes \sigma_z, X]$, and the perturbation is $\mathcal{V}_{\mathrm{pert}} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. Because $\mathrm{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space).

$$(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})X = \sigma_z \otimes \Big(-ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big) \in \mathcal{H}_F. \tag{69}$$

Step 2 (Poisson equation on the fast ancilla). In finite dimensions, the Moore–Penrose pseudoinverse can be written as

$$\mathcal{R}^+ := -\int_0^\infty \left(e^{s\mathcal{R}} - P_S\right) ds, \tag{70}$$

so that $\mathcal{R}\mathcal{R}^+ = \mathrm{id} - P_S = P_F$ and $P_S\mathcal{R}^+ = \mathcal{R}^+P_S = 0$. In particular, since $\mathcal{R}_F(\sigma_z) = -\kappa\,\sigma_z$ with $\kappa := 4\Gamma$, we have $\mathcal{R}^+(\sigma_z \otimes Y) = \mathcal{R}_F^+(\sigma_z) \otimes Y = -(1/\kappa)\sigma_z \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})X = -\frac{1}{\kappa}\sigma_z \otimes \Big(-ig\,[\sigma_z, X_S] + \eta\,\mathcal{G}_S(X_S)\Big). \tag{71}$$

Step 3 (return map and projection). Since $\mathcal{V}_{\mathrm{skew}}(\sigma_z \otimes Y) = -ig\,\mathbf{1}_F \otimes [\sigma_z, Y]$ and $\mathcal{V}_{\mathrm{pert}}(\sigma_z \otimes Y) = \mathbf{1}_F \otimes \mathcal{G}_S(Y)$, we compute

$$(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})\,\mathcal{R}^+(\mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}})X \tag{72}$$

$$= \frac{1}{\kappa}\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z, [\sigma_z, X_S]] + i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] + i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) - \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{73}$$

Projecting with $P_S(\mathbf{1}_F \otimes Y) = \mathbf{1}_F \otimes Y$ yields

$$\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(g^2\,[\sigma_z, [\sigma_z, X_S]] \tag{74}$$

$$+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \tag{75}$$

$$+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \tag{76}$$

$$- \eta^2\,\mathcal{G}_S^2(X_S)\Big). \tag{77}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z, [\sigma_z, X_S]]. \tag{78}$$

Non-self-adjointness on the slow space. Since $\pi_S \propto \mathbf{1}_S$, the $L^2(\pi_S)$ inner product is proportional to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}^\dagger[\sigma_+] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $L^2(\pi_S)$. Consequently, the $\mathcal{O}(\eta)$ mixed operator

$$\mathcal{B} := ig\,\big(\,\mathrm{ad}_{\sigma_z} \circ \mathcal{G}_S + \mathcal{G}_S \circ \mathrm{ad}_{\sigma_z}\,\big), \qquad \mathrm{ad}_{\sigma_z}(X) := [\sigma_z, X], \tag{79}$$

is generically not self-adjoint either (unless additional commutation/self-adjointness relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$.

Verification of Assumption 3.1. All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 3.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 3.1.2 holds since both $\mathcal{V}_{\rm skew}$ and $\mathcal{V}_{\rm pert}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 3.1.3 holds with $M_R := \|\mathcal{R}\|$. Moreover, in this toy model the flux regularity bound (11) can be checked explicitly. For $\psi = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$ one has $g_\psi = P_F \mathcal{V}_{\rm skew} P_S \psi = -ig\,\sigma_z \otimes [\sigma_z, X_S]$ and $(-\mathcal{R}^+)g_\psi = (1/\kappa)g_\psi$. Therefore

$$P_F \mathcal{V}_{\rm skew} P_F (-\mathcal{R}^+) g_\psi = 0, \tag{80}$$

$$(-Q_O)\psi = P_S \mathcal{V}_{\rm skew} P_F (-\mathcal{R}^+) g_\psi = \frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z, [\sigma_z, X_S]]. \tag{81}$$

Since the left-hand side is identically zero, (11) holds (e.g. with $\kappa_{\rm ell} := 1$).
(4) For Assumption 3.1.4, decompose $\mathcal{V}_{\rm pert} = \mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$ by $\mathcal{K}_{\rm pert} := \frac{1}{2}(\mathcal{V}_{\rm pert} - \mathcal{V}_{\rm pert}^\dagger)$ and $\mathcal{S}_{\rm pert} := \frac{1}{2}(\mathcal{V}_{\rm pert} + \mathcal{V}_{\rm pert}^\dagger)$. Since $\mathcal{V}_{\rm pert}$ is bounded, the energy bounds (14)–(15) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\rm pert}\|$, $b_2 = \|\mathcal{S}_{\rm pert}\|$.

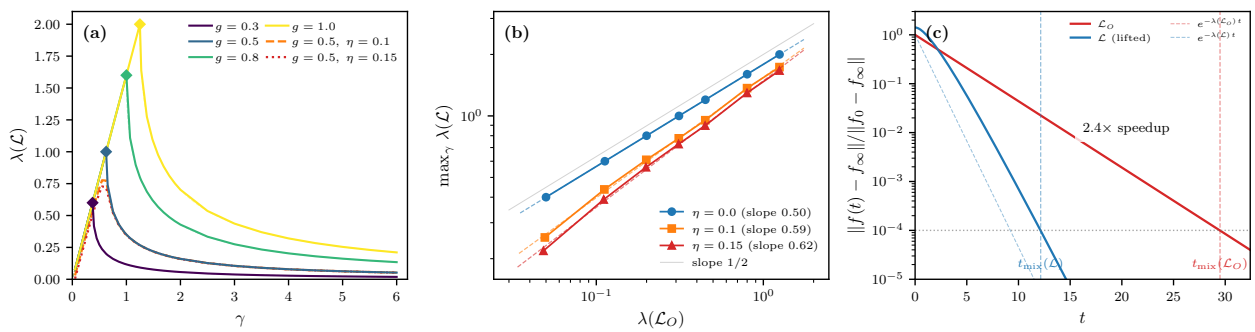

FIGURE 3. Numerical verification for Example 5.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the exact analytic formula (82), diamond markers indicate the exact peaks. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(\mathcal{L}_O)$ for $\eta \in \{0, 0.10, 0.15\}$; fitted slopes are shown in the legend. **(c)** Semigroup decay for the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes $e^{-\lambda(\mathcal{L})\,t}$ and $e^{-\lambda(\mathcal{L}_O)\,t}$. Vertical markers indicate the ε-mixing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.2, \ldots, 1.0\}$.

Numerical verification. We assemble the exact 16×16 Liouville generator and compute spectral gaps. The scan uses $g \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8, 1.0\}$, $\Gamma = 0.8$ (so $\kappa = 3.2$), $\gamma \in [0.05, 6.0]$ (38 points), $\eta \in \{0, 0.10, 0.15\}$. For $\eta = 0$, there is an exact formula for the slow-sector eigenvalues of $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\rm skew}$:

$$\lambda_\pm = \tfrac{1}{2}\big(-\gamma\kappa \pm \sqrt{(\gamma\kappa)^2 - 16g^2}\big). \tag{82}$$

This formula gives the three regimes directly: underdamped ($\gamma < \gamma^* = g/\Gamma$), critical ($\gamma = \gamma^*$), and overdamped ($\gamma > \gamma^*$ with asymptotic gap $g^2/(\Gamma\gamma)$). Numerically, agreement with (82) is at machine precision. The optimizer $\gamma^* = g/\Gamma$ is consistent with the scaling law $\gamma_{\rm opt} \asymp \sqrt{\lambda(\mathcal{L}_O)}$ up to a model-dependent constant.

Panel (b) shows $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$ with slope near 1/2, i.e. $\max_\gamma \lambda(\mathcal{L}) \propto \sqrt{\lambda(\mathcal{L}_O)}$. For $\eta \neq 0$, one slow eigenvalue of $\mathcal{L}_O$ becomes unstable at order η^2/κ; therefore we report the stable gap $\lambda_{\rm stab}(\mathcal{L}) := \min\{|\Re\mu| : \Re\mu < 0\}$. Using this standard convention, the scaling trend is unchanged. Panel (c) shows a representative mixing-time gain: for $g = 0.5$, $\eta = 0$, $\varepsilon = 10^{-4}$, $t_{\rm mix}(\mathcal{L}) \approx 12.2$ and $t_{\rm mix}(\mathcal{L}_O) \approx 29.5$. The smaller gain in mixing time than in gap is due to transient non-normal effects in the lifted dynamics. At $\eta = 0.1$, we also verify $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \sim 10^{-15}$.

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as above, now in the boundary-driven toy model.

Step 1 (compute slow operators). Compute $\mathcal{L}_O$ and Q_O from the physical model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\lambda_R\,\mathrm{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. This produces an 8×8 lifted generator (the physical model is 16×16).

Step 3 (match the non-equilibrium part). Solve the consistency relation (59) numerically for the perturbation block. This linear algebra problem has multiple solutions; we choose the one with smallest matrix norm. The final residual is below 10^{-12}.

Figure 4 compares algorithmic and physical lifts on the matched grid $g \in \{0.3, 0.5, 0.8, 1.0\}$ and $\eta \in \{0.10, 0.15\}$. The peak gaps are identical (Table 2), and the scaling in Panels (a)–(b) is the same for both constructions. Panel (c) shows nearly identical decay curves. For the steady state, at $g = 0.5$, $\eta = 0.1$ we get $\|P_S h_\eta^{\mathrm{alg}} - g_\eta\|/\|g_\eta\| \lesssim 10^{-12}$. So in this model, the generic lift reproduces the same accelerated behavior as the physical lift.

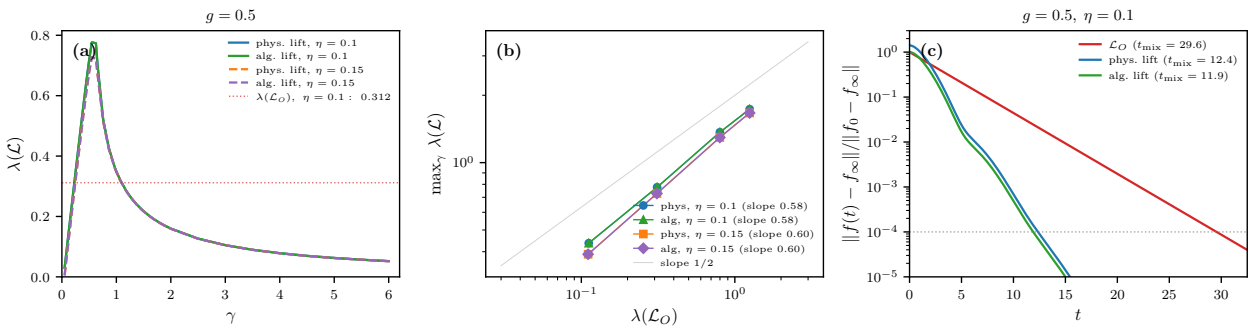

FIGURE 4. Comparison of the *physical* (16×16) and *algorithmic* (8×8, square-root factorization) lifted generators for Example 5.3. **(a)** Spectral gap vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped gap $\lambda(\mathcal{L}_O)$ is shown as a horizontal reference. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$ for both lifts; fitted slope is close to 1/2. **(c)** Mixing-time comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 2. Peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts. All entries agree to at least 4 significant digits (maximum relative difference $<10^{-10}$). The "speedup" column is $\max_\gamma \lambda(\mathcal{L})/\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	speedup
0.3	0.10	0.1117	0.4366	0.4366	3.91×
0.3	0.15	0.1107	0.3897	0.3897	3.52×
0.5	0.10	0.3117	0.7766	0.7766	2.49×
0.5	0.15	0.3107	0.7279	0.7279	2.34×
0.8	0.10	0.7992	1.3674	1.3674	1.71×
0.8	0.15	0.7982	1.2954	1.2954	1.62×
1.0	0.10	1.2492	1.7335	1.7335	1.39×
1.0	0.15	1.2482	1.6671	1.6671	1.34×

Remark 5.6 (Bias behaviour in the two examples). Both examples show extremely small projection bias, but with different strength of justification. In the boundary-driven model, we prove exact zero

bias: $\gamma S(\gamma) = \mathcal{L}_O$ (because slow–fast excursions stay in the σ_z channel, where the effective fast–fast feedback vanishes in the Schur map), and therefore $P_S h_\eta = g_\eta$ for all γ. In the Langevin model, we obtain numerical evidence of the same behavior: $\|P_S h_\eta - g_\eta\|/\|g_\eta\| \lesssim 10^{-14}$ at $\eta = 0.1$. Hence no accuracy floor is observed in either experiment. For models with $P_F \mathcal{V} P_F \neq 0$, use the two-phase strategy of Remark 5.3.

6. Conclusion and open problems

We have developed a quantitative hypocoercive framework for generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on a slow–fast Hilbert decomposition, providing an explicit characterisation of the interplay between the friction γ, the reversible backbone gap $\lambda(Q_O)$, and the non-equilibrium driving strength η. The main conclusions are: First, Theorem 4.1 gives an explicit η-dependent lower bound $\lambda_\eta \geq \frac{2}{3}(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma))$, so deterioration in η is linear with computable slope. Second, Corollary 4.2 yields the stability scale $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ at optimal friction: faster reversible mixing allows stronger forcing. Third, Theorem 4.4 gives an overdamped ceiling, $\mathcal{O}(\gamma^{-1}\lambda(\mathcal{L}_O)) + \mathcal{O}(\eta)$, confirming that acceleration is a finite-friction effect.

Within the stable regime, $\lambda(\mathcal{L}_O) \approx \lambda(Q_O)$ (Remark 3.4), so either gap can be used consistently to describe scaling. For designed lifts built by reverse adiabatic elimination (Section 5.1), Proposition 5.1 shows that imposing $P_F \mathcal{V}_{\text{pert}} P_F = 0$ removes the projection bias ($\varepsilon_* = 0$), giving full quadratic speedup for any target accuracy. The boundary-driven example proves this exactly; the Langevin example shows the same behavior numerically up to machine precision (Remark 5.6). For natural models with $P_F \mathcal{V} P_F \neq 0$, the two-phase extension in Remark 5.3 remains available.

We conclude with several directions for future investigation.

(1) Sharp stability thresholds. The threshold η^* in Theorem 4.1 is derived from a Lyapunov argument and involves structural constants that may not be sharp. Numerically, the accelerated regime appears to persist well beyond η^*; sharpening this threshold is an important open problem. In particular, for models where the non-reversible perturbation *enhances* the effective gap (e.g. optimally rotated diffusions [10]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a refined analysis incorporating this enhancement could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{83}$$

(2) Optimal factorization for the reverse adiabatic elimination. The reverse adiabatic elimination recipe (Section 5.1) shows that any target Q_O can be lifted, but the resulting structural constants—and hence the stability threshold—depend on the choice of auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the factorization that maximises η^* (and minimises K_{geom}, κ_{ell}) in infinite-dimensional settings where domain issues (Remark 3.2) constrain the couplings remains open. In classical sampling, multilevel ("replica") methods and cascaded Langevin lifts [19] have been shown to provide iterated acceleration. Extending the present framework—including the non-equilibrium stability analysis—to a chain of nested slow–fast decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$ could yield explicit multilevel stability thresholds and rate bounds.

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Appendix A. Proofs for Preliminaries

This appendix contains the proofs of the spectral bounds and hypocoercivity characterizations stated in Section 2. The arguments are carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$. In the noncommutative realization, this corresponds to $\mathcal{H} = L^2(\sigma)$ and $\mathcal{H}_0 = \{X : \mathrm{tr}(\sigma X) = 0\}$.

A.1. **Proof of Lemma 2.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part* $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$*, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{84}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{85}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy-Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma 2.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

(⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \leq -\lambda_S \|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{86}$$

For any $X_t = \mathcal{P}_t X_0 \in \mathcal{H}_0$, the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \leq -\lambda_S \|X_t\|^2. \tag{87}$$

Integrating this differential inequality yields $\|X_t\| \leq e^{-\lambda_S t}\|X_0\|$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

(⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \leq e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. The Hille-Yosida theorem (or simple differentiation at $t = 0$) implies that the generator satisfies the dissipativity condition:

$$\Re\langle X, \mathcal{L}X\rangle \leq -\nu\|X\|^2, \quad \forall X \in \mathcal{H}_0. \tag{88}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \leq -\nu\|X\|^2. \tag{89}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \leq -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B. Proof of Lemma 3.8

This appendix establishes the structural bounds of Lemma 3.8.

Proof. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. By Assumption 3.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \leq \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{90}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (9), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\text{skew}}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{91}$$

Let $v := \mathcal{V}_{\text{skew}}u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (90):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{92}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{93}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{94}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{95}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\mathrm{dom}(Q_O) = \mathcal{H}_S$, so $u \in \mathrm{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\mathrm{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \mathrm{dom}(Q_O)$; we may therefore apply the flux conditioning bound (9) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{96}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}}\varpi^{-1/2}$ and $\|\mathcal{V}_{\mathrm{skew}}\mathcal{A}_\varpi\| \le k_{\mathrm{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\mathrm{reg}} := \sqrt{M_R/\lambda_R}. \tag{97}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\varpi\mathcal{V}_{\mathrm{skew}}P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\varpi^\dagger = \left((\varpi - Q_O)^{-1}P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}P_S(\varpi - Q_O)^{-1}. \tag{98}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\mathrm{skew}}^\dagger = -\mathcal{V}_{\mathrm{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F\mathcal{V}_{\mathrm{skew}}^\dagger\mathcal{A}_\varpi^\dagger = P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)\mathcal{V}_{\mathrm{skew}}P_S(\varpi - Q_O)^{-1}. \tag{99}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1}g$. Define the macroscopic flux

$$g_\psi := P_F\mathcal{V}_{\mathrm{skew}}P_S\psi \in \mathcal{H}_F. \tag{100}$$

Then $\mathcal{T}^\dagger g = P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\psi$. Invoking the **Flux Regularity** condition from Assumption 3.1.3:

$$\|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\mathrm{ell}}\|(-Q_O)\psi\|. \tag{101}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\mathrm{ell}}\|(-Q_O)(\varpi - Q_O)^{-1}g\|. \tag{102}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\varpi - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\mathrm{ell}}\|g\|. \tag{103}$$

This proves $\|\mathcal{A}_\varpi\mathcal{V}_{\mathrm{skew}}P_F\| \le \kappa_{\mathrm{ell}}$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ from Assumption 3.1.4. We also recall the centered mapping property (31), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 3.1.4(a)):

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{104}$$

For the symmetric part, we use the energy bound (Assumption 3.1.4(b)):

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in \operatorname{dom}((-\mathcal{R})^{1/2}). \tag{105}$$

Reverse estimate ($\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\text{pert}}^\dagger = -\mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le \|\mathcal{K}_{\text{pert}} u\| + \|\mathcal{S}_{\text{pert}} u\| \tag{106}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{107}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\text{pert}}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{108}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \varpi^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi f\| \le (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2} \|f\|. \tag{109}$$

This establishes (28a) with $k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}$.

Forward estimate ($\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\varpi \mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^\dagger\| = \|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\|\mathcal{V}_{\text{pert}}^\dagger w\| \le \|\mathcal{K}_{\text{pert}} w\| + \|\mathcal{S}_{\text{pert}} w\|$$

$$\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \tag{110}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\| \|g\| \le k_{\mathcal{A}} \varpi^{-1/2}. \tag{111}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\text{skew}} P_S (\varpi - Q_O)^{-1} g\|. \tag{112}$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi, \tag{113}$$

$$\|(-\mathcal{R})^{-1/2} g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2, \tag{114}$$

hence

$$\|(-\mathcal{R})^{1/2} w\| = \|(-Q_O)^{1/2} (\varpi - Q_O)^{-1} g\| \le \frac{1}{2} \varpi^{-1/2} \|g\|. \tag{115}$$

(The last step follows from the scalar bound $\sup_{x \ge 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (110) yields

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\varpi}^{\dagger}g\| \leq \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\varpi^{-1/2}\,\|g\|. \tag{116}$$

Since g is arbitrary, this establishes (28b) with $\tilde{k}_P(\varpi) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi^{-1/2}$.

□

Appendix C. Proof of Theorem 4.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_{\varpi}f, f\rangle, \tag{117}$$

where $\mathcal{A}_{\varpi} = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

C.1. **Preliminaries.** We strictly utilize the structural estimates established in Lemma 3.8. We recall the constants derived therein, depending solely on the parameters from Assumption 3.1 $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$. For convenience in the Lyapunov dissipation bounds below, we introduce the effective symmetric-perturbation coefficient

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{118}$$

which absorbs both the slow-mode contribution $b_2 + \frac{a_2}{2}$ and the fast-mode λ_R-amplification $\frac{a_2\lambda_R}{2}$ that arises below when applying $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ to the symmetric perturbation energy. The relevant bounds from Lemma 3.8 are the Solver Bound $\|\mathcal{A}_{\varpi}\| \leq k_{\mathcal{A}}\varpi^{-1/2}$ with $k_{\mathcal{A}} := (2\sqrt{\lambda_R})^{-1}$, the Regularity estimates $\|\mathcal{V}_{\text{skew}}\mathcal{A}_{\varpi}\| \leq k_{\text{reg}} := \sqrt{M_R/\lambda_R}$ and $\|\mathcal{A}_{\varpi}\mathcal{V}_{\text{skew}}P_F\| \leq \kappa_{\text{ell}}$, and the Perturbation Control bounds

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\varpi}\| \leq k_P(\varpi) = (b_1+b_2)k_{\mathcal{A}}\,\varpi^{-1/2}, \tag{119a}$$

$$\|\mathcal{A}_{\varpi}\mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\varpi) = \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}. \tag{119b}$$

We denote the spectral gap of the self-adjoint operator $-Q_O$ by $\lambda(Q_O)$.

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the computations below are performed on a common invariant core $\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}(\mathcal{A}_{\varpi})$ and extended to the energy domain by density.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \text{dom}(\mathcal{R})$,*

$$\Re\langle\mathcal{R}f, f\rangle = \langle\mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{120}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle\mathcal{R}f, f\rangle = \Re\langle\mathcal{R}f_F, f_F\rangle. \tag{121}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle\mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t satisfy $\partial_t f_t = \mathcal{L} f_t$. By differentiating (33) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \partial_t f_t\rangle \tag{122}$$

$$= \Re\langle \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \mathcal{L} f_t\rangle. \tag{123}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{124}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L} f, f\rangle, \tag{125}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{126}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R} f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{127}$$

Using Assumption 3.1.1 ($\mathcal{R}$ coercivity) and Assumption 3.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R} f, f\rangle \le -\gamma\lambda_R \|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (15) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{128}$$

$$\le \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{129}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R} f, f\rangle = \Re\langle \mathcal{R} f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Hence

$$\gamma\Re\langle \mathcal{R} f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{130}$$

$$\le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{131}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{132}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{133}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{134}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle, \tag{135}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{skew}} f\rangle, \tag{136}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\mathrm{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\mathrm{pert}} f\rangle. \tag{137}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle. \tag{138}$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\varpi f, f\rangle = 0. \tag{139}$$

Moreover $\mathcal{A}_\varpi \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f_F, f_S\rangle. \tag{140}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\mathrm{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi \mathcal{R} f_F = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)\mathcal{R} f_F = -(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F. \tag{141}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\text{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\left|\Re\langle (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F, f_S\rangle\right| = \alpha\gamma\left|\Re\langle f_F, \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\rangle\right| \tag{142}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|. \tag{143}$$

To bound $\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{144}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\text{skew}} \psi = 0$, hence $g_\psi = \mathcal{V}_{\text{skew}}\psi = \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (9) yields

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{145}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\text{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\text{skew}}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{146}$$

Therefore

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \sqrt{M_R}\,\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} f_S\|. \tag{147}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{x\geq 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{148}$$

which implies

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_S\|. \tag{149}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{150}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle, \tag{151}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (23)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{152}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_F\rangle. \tag{153}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle. \tag{154}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\rm skew} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{155}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{156}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{157}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\rm skew} P_S = 0$, we have $P_F \mathcal{V}_{\rm skew} f_S = \mathcal{V}_{\rm skew} f_S$. By the definition of $\mathcal{A}_\varpi$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle = \alpha\langle (\varpi - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\varpi - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{158}$$

$$\le -\alpha\,\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}\,\|f_S\|^2. \tag{159}$$

Cross Terms. Using the Regularity estimates:

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle + \alpha\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle \tag{160}$$

$$\le \alpha(\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\| + \|\mathcal{V}_{\rm skew} \mathcal{A}_\varpi\|)\|f_F\|\|f_S\| \tag{161}$$

$$\le \alpha(\kappa_{\rm ell} + k_{\rm reg})\|f_F\|\|f_S\|. \tag{162}$$

Fast–fast remainder. Using $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\rm skew}\mathcal{A}_\varpi f_F, f_F\rangle\big| \le \alpha\,\|\mathcal{V}_{\rm skew}\mathcal{A}_\varpi\|\,\|f_F\|^2 \le \alpha\,k_{\rm reg}\,\|f_F\|^2. \tag{163}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle. \tag{164}$$

We estimate each term using Lemma 3.8 and the centered mapping property (31).
Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\rm pert}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle\big| \le \alpha\eta\,\|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f\|\,\|f_F\| \le \alpha\eta\,k_P(\varpi)\,\|f\|\,\|f_F\| \tag{165}$$

$$\le \alpha\eta\,k_P(\varpi)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{166}$$

Forward term. Since $\mathcal{A}_\varpi \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle\big| \le \alpha\eta\,\|\mathcal{A}_\varpi \mathcal{V}_{\rm pert} f\|\,\|f_S\| \le \alpha\eta\,\tilde{k}_P(\varpi)\,\|f\|\,\|f_S\| \tag{167}$$

$$\le \alpha\eta\,\tilde{k}_P(\varpi)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{168}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle Xf_F, f_S\rangle| \le \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) \le -\mathbf{v}(t)^T\mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix}\|P_F f_t\| \\ \|P_S f_t\|\end{pmatrix}. \tag{169}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α,ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big) - \alpha k_{\mathrm{reg}} - \alpha\eta k_P(\varpi) \\ &= \gamma\lambda_R - \eta b_{2,\mathrm{eff}} - \alpha k_{\mathrm{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\varpi^{-1/2}. \end{aligned} \tag{170}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2+\tfrac{a_2}{2}\big) - \alpha\eta\,\tilde{k}_P(\varpi) \\ &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2+\tfrac{a_2}{2}\big) - \alpha\eta\left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}\right]. \end{aligned} \tag{171}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2} + (k_{\mathrm{reg}}+\kappa_{\mathrm{ell}}) + \eta\big(k_P(\varpi)+\tilde{k}_P(\varpi)\big)\right] \\ &= -\frac{1}{2}\alpha\left[k_{\mathrm{reg}}+\kappa_{\mathrm{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\varpi^{-1/2}\right]. \end{aligned} \tag{172}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)}. \tag{173}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\varpi+\lambda(Q_O))$, while the cross term $|M_{12}^0|$ contains $\gamma\Theta\varpi^{-1/2}$. A leading-order balance (maximising $\varpi(\varpi+\lambda(Q_O))^{-2}$ in the large-γ rate) selects

$$\varpi_* := \lambda(Q_O). \tag{174}$$

At this value, $\lambda(Q_O)/(\varpi_*+\lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\mathrm{geom}}, Z(\gamma)$ from (35)–(36), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \tag{175}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{176}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\,Z(\gamma). \tag{177}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\mathrm{reg}} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{178}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{179}$$

in agreement with (37), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\text{reg}}}. \tag{180}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\text{reg}}) + \frac{1}{2}\alpha_*$. Dropping the non-positive term $-\alpha_* k_{\text{reg}}$ gives the conservative upper bound

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \le \gamma\lambda_R + \tfrac{1}{2}\alpha_* = \gamma\lambda_R\left(1 + \frac{\frac{1}{2}}{Z(\gamma)^2 + 2k_{\text{reg}}}\right). \tag{181}$$

Applying (173) yields

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \ge \frac{\frac{1}{4}\,\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}} =: \Lambda_{\text{hypo}}(\gamma), \tag{182}$$

consistent with definition (38). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (182) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{183}$$

Thus, $\Lambda_{\text{hypo}}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants K_{geom}, $2k_{\text{reg}}$, and $\frac{1}{2}$. These arise from the slow–fast coupling through $\mathcal{V}_{\text{skew}}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\text{hypo}}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\text{eff}} + \alpha_*\, k_P(\varpi_*), \tag{184}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\, \tilde{k}_P(\varpi_*), \tag{185}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{186}$$

Here E_{11} contains the full effective coefficient $b_{2,\text{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\text{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2}f\|^2 \ge \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (132)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \le \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{187}$$

Since $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ (as $\lambda_R \geq 0$), we have $E_{11} \geq E_{22}$, so the maximum is attained by the first row. To obtain a uniform closed-form upper bound, we additionally replace k_P with $\tilde{k}_P$ (since $\tilde{k}_P \geq k_P$) in E_{11}:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\left[\tfrac{1}{2}\,\tilde{k}_P(\varpi_*) + \tfrac{3}{2}\,\tilde{k}_P(\varpi_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\varpi_*). \tag{188}$$

Recall from Lemma 3.8 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + \frac{3}{4}(a_1 + a_2). \tag{189}$$

Indeed, $2\tilde{k}_P(\varpi_*) = 2\big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi_*^{-1/2}$; using $\frac{1}{2}k_P + \frac{3}{2}\tilde{k}_P = (2(b_1+b_2)k_{\mathcal{A}} + \frac{3(a_1+a_2)}{4})\varpi^{-1/2} = C_{\text{sing}}\varpi^{-1/2}$ (tighter than $2\tilde{k}_P$), the bound (188) refines to

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\varpi_*^{-1/2}. \tag{190}$$

Substituting $\alpha_*(\gamma)$ from (37), the bound becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{191}$$

consistent with definition (39).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which defines the threshold

$$\eta^*(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|} \geq \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)}. \tag{192}$$

Substituting the explicit forms of Λ_{hypo} (38) and B_{pert} (39) yields

$$\eta^*(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}\big)\big(2b_{2,\text{eff}} + 2\alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\big)}. \tag{193}$$

Asymptotic analysis of stability. We analyze the scaling of $\eta^*(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.
Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta^*(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta^*(\gamma) \asymp \mathcal{O}(1)$.
Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{194}$$

The singularity thus cancels, and at the optimal friction the stability threshold is well-behaved:

$$\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{195}$$

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter $\delta(\gamma)$ defined by

$$\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{196}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}$, we obtain the crude but uniform estimate

$$\delta(\gamma) \le 2\,\frac{\gamma\lambda_R}{Z(\gamma)^2}\, k_{\mathcal{A}}\, \varpi_*^{-1/2} = \frac{\gamma\sqrt{\lambda_R}\,\varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\mathrm{geom}}\big)^2}. \tag{197}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta(\gamma) \le \frac{\sqrt{\lambda_R}}{\Theta}\,\frac{x}{(x+K_{\mathrm{geom}})^2} \le \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\mathrm{geom}}} =: \delta_{\max}. \tag{198}$$

Since $K_{\mathrm{geom}} \ge \sqrt{M_R/\lambda_R}$ and $\Theta = \sqrt{M_R}/2$, we obtain $\delta_{\max} \le \frac{\lambda_R}{2M_R}$. By Assumption 3.1.3, the flux conditioning constant M_R is monotone (any larger value also satisfies the assumption); we may therefore take $M_R \ge \lambda_R$ without loss of generality, which gives $\delta_{\max} \le \frac{1}{2}$. (Choosing $M_R > \lambda_R$ can only *decrease* $\delta_{\max} = \lambda_R/(2M_R)$ relative to its natural value $\lambda_R/(2M_R^\natural)$ where $M_R^\natural$ is the smallest valid constant; when $M_R^\natural \ge \lambda_R$ already, no enlargement is needed. The assumption $M_R \ge \lambda_R$ is verified in both examples of Section 5.) This implies that the condition number of the norm equivalence is uniformly bounded for all γ:

$$C_\gamma := \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}} \le \sqrt{\frac{1+1/2}{1-1/2}} \tag{199}$$

$$= \sqrt{3}. \tag{200}$$

Thus, we have the uniform equivalence bounds:

$$\frac{1}{4}\|f\|^2 \le \frac{1}{2}(1-\delta(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\|f\|^2 \le \frac{3}{4}\|f\|^2. \tag{201}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{202}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{203}$$

Next we use the norm equivalence from Proposition 3.10 in the convenient form

$$\frac{1}{2}(1-\delta(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\,\|f\|^2, \tag{204}$$

where $\delta(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\varpi_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f). \tag{205}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t). \tag{206}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\varpi_*}(f_0). \tag{207}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \frac{1+\delta(\gamma)}{1-\delta(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}t}\,\|f_0\|^2. \tag{208}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}}\, e^{-\lambda_\eta t}\, \|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta(\gamma)}. \tag{209}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{210}$$

For the perturbation norm, we use the sharp bound (190) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}}_{\alpha_*(\gamma)}\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}. \tag{211}$$

Substituting these into the expression for λ_η and using the uniform bound $1+\delta(\gamma) \le 3/2$:

$$\lambda_\eta \ge \frac{2}{3}\left[\frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}} - \eta\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}\right)\right]. \tag{212}$$

This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/Z(\gamma)^2$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}} f\|_{\mathcal{H}} \le \sqrt{3}\, e^{-\lambda_\eta t} \|f\|_{\mathcal{H}}. \tag{213}$$

This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine γ_{opt} maximising $\Lambda_{\text{hypo}}(\gamma)$. From (182), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)} \cdot \frac{x}{(x+K_{\text{geom}})^2 + \left(2k_{\text{reg}} + \frac{1}{2}\right)}. \tag{214}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\text{geom}})^2 + C_0]$ with $C_0 := 2k_{\text{reg}} + \frac{1}{2}$ is maximised when $(x+K_{\text{geom}})^2 + C_0 = 2x(x+K_{\text{geom}})$, equivalently $x^2 = K_{\text{geom}}^2 + C_0$. Translating back:

$$\gamma_{\text{opt}} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\text{lift}}\sqrt{\lambda(Q_O)}, \tag{215}$$

where $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}$.

Maximal rate. At $\gamma = \gamma_{\text{opt}}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\text{opt}} = \sqrt{K_{\text{geom}}^2 + C_0}$. Substituting into the denominator: $(x_* + K_{\text{geom}})^2 + C_0 = 2(K_{\text{geom}}^2 + C_0 + K_{\text{geom}}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_* + K_{\text{geom}})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{216}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Stability threshold at optimal lift. The admissible perturbation range $\eta^*(\gamma_{\rm opt})$ at the accelerated regime is determined as follows. From (192), the threshold depends on $\|\mathbf{E}(\gamma)\|$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at $\gamma_{\rm opt}$:

$$\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2 + 2k_{\rm reg}}. \tag{217}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2 + 2k_{\rm reg} + \frac{1}{2}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2 + 2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt}) \cdot \lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (190):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{218}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $\|\mathbf{E}(\gamma_{\rm opt})\| = \mathcal{O}(1)$, the stability threshold scales as

$$\eta^*(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2\|\mathbf{E}(\gamma_{\rm opt})\|} \asymp \sqrt{\lambda(Q_O)}. \tag{219}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 4.4.

D.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{220}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{221}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le C\, s(\mathcal{L}). \tag{222}$$

Proof. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f\, dt. \tag{223}$$

The assumed exponential bound $\|e^{t\mathcal{L}} f\| \le Ce^{-\nu t}\|f\|$ implies that the Bochner integral converges in $\mathcal{H}_0$ and

$$\|u\| \le \int_0^\infty \|e^{t\mathcal{L}} f\|\, dt \le \int_0^\infty Ce^{-\nu t}\, dt\, \|f\| = \frac{C}{\nu}\, \|f\|. \tag{224}$$

Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L}e^{t\mathcal{L}} f\, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big)\, dt = f. \tag{225}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le C/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/C$, i.e. $\nu \le C\, s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume there exist constants* $C \geq 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{226}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \geq 0\,;\ \|e^{t\mathcal{L}} f\| \leq e^{-1}\, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{227}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \geq \frac{1}{2C\, s(\mathcal{L})}. \tag{228}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [4].

Proof. By the assumed exponential bound $\|e^{t\mathcal{L}}\| \leq Ce^{-\nu t}$, for any $r \geq 0$ we have $\|e^{r\mathcal{L}}\| \leq C$. By the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for all $t \geq 0$ we can write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}_0$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$, and obtain from the semigroup property and $\|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\| \leq e^{-1}$:

$$\|e^{t\mathcal{L}}\| \leq \|e^{r\mathcal{L}}\|\, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \leq C\, e^{-n}.$$

As in the proof of Lemma D.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}}\, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \leq \int_0^\infty \|e^{t\mathcal{L}}\|\, dt \leq \sum_{n=0}^{\infty} \int_0^{t_{\mathrm{rel}}(\mathcal{L})} C\, e^{-n}\, dr = \frac{C\, t_{\mathrm{rel}}(\mathcal{L})}{1 - e^{-1}} \leq 2C\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \geq (2C\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 4.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{229}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} \phi \in \mathcal{H}_F, \tag{230}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \geq 1. \tag{231}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = P_F \mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{232}$$

Consequently,

$$\mathcal{L} f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{233}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \ +\ \eta\mathcal{V}_{\mathrm{pert}} f_\gamma. \tag{234}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}}\psi + P_F \mathcal{V}_{\mathrm{skew}}\psi. \tag{235}$$

For the slow part, using $Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S$ and $P_S\phi = \phi$, we have

$$P_S \mathcal{V}_{\mathrm{skew}}\psi = P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{236}$$

For the fast part, apply the flux regularity estimate (11) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\rm skew}P_S\phi = \mathcal{V}_{\rm skew}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\rm skew}\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F\psi\| = \|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\rm ell}\,\|(-Q_O)\phi\| = \kappa_{\rm ell}\,\lambda(Q_O). \tag{237}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\rm skew}\psi\| \le (1+\kappa_{\rm ell})\,\lambda(Q_O). \tag{238}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\rm pert}f_\gamma\|. \tag{239}$$

To bound the perturbation, use Assumption 3.1.4 to estimate

$$\|\mathcal{V}_{\rm pert}f\| \le \|\mathcal{K}_{\rm pert}f\| + \|\mathcal{S}_{\rm pert}f\| \tag{240}$$

$$\le (a_1+a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1+b_2)\,\|f\|, \qquad f\in{\rm dom}((-\mathcal{R})^{1/2}). \tag{241}$$

Since $\phi\in\ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{242}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi,\psi\rangle = \langle\mathcal{V}_{\rm skew}\phi,(-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi,\phi\rangle = \lambda(Q_O). \tag{243}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert}f_\gamma\| \le (b_1+b_2)\,\|f_\gamma\| + (a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{244}$$

Dividing by $\|f_\gamma\|\ge 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \le \frac{(1+\kappa_{\rm ell})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{245}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 5.5

We verify Assumption 3.1.1–4 for the forced Langevin decomposition on the core $\mathcal{C}_{\rm Lan} := C^\infty(\mathbb{T}^d)\otimes\mathcal{S}(\mathbb{R}^d)\subset L^2(\mu_0)$.

E.1. **Microscopic spectral gap and centering.** The Ornstein–Uhlenbeck operator $\mathcal{R} = -p\cdot\nabla_p + \beta^{-1}\Delta_p$ acts only on p-variables and has spectral gap $\lambda_R = 1$ on $\mathcal{H}_F = \ker(\Pi_p)^\perp$ (classical; see e.g. [24]). For Assumption 3.1.2: $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ (Hamiltonian flow preserves μ_0), $P_S\mathcal{V}_{\rm skew}P_S = 0$ (Gaussian centering $\int p\,d\nu_p = 0$), and $P_S\mathcal{V}_{\rm pert}P_S = 0$ (since $\nabla_p\phi(q) = 0$).

E.2. **Flux regularity (Assumption 3.1.3).** For $\psi = \psi(q)\in\mathcal{H}_S$, the flux is $g_\psi = p\cdot\nabla_q\psi\in\mathcal{H}_F$. Since $\mathcal{R}(p_i) = -p_i$, we have $(-\mathcal{R}^+)g_\psi = g_\psi$, giving $M_R = 1$.

For the elliptic regularity bound, define $Z_{ij}(p) := p_ip_j - \beta^{-1}\delta_{ij}$. A direct computation yields

$$P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi = \sum_{i,j} Z_{ij}(p)\,\partial_{q_iq_j}\psi. \tag{246}$$

By Cauchy–Schwarz and Gaussian moment calculations ($\mathbb{E}[p_i^4] = 3\beta^{-2}$),

$$\|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi\|_{L^2(\mu_0)} \le c_{d,\beta}\,\|\nabla_q^2\psi\|_{L^2(\nu)}, \qquad c_{d,\beta} = \frac{\sqrt{d(d+1)}}{\beta}. \tag{247}$$

It remains to control $\|\nabla_q^2\psi\|_{L^2(\nu)}$ by $\|(-Q_O)\psi\|_{L^2(\nu)}$.

Step 1 (Hessian vs. Laplacian). By Parseval's identity on $\mathbb{T}^d$, $\|\nabla_q^2\psi\|_{L^2(dq)} = \|\Delta_q\psi\|_{L^2(dq)}$ for periodic ψ. Switching to $L^2(\nu)$ with $\rho_{\min} \le \rho \le \rho_{\max}$ ($\rho \propto e^{-\beta U}$) gives

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le \sqrt{\rho_{\max}/\rho_{\min}}\,\|\Delta_q\psi\|_{L^2(\nu)}. \tag{248}$$

Step 2 (Laplacian vs. Q_O). From $\Delta_q\psi = \beta(Q_O\psi + \nabla U\cdot\nabla_q\psi)$ and the Dirichlet-form identity $\langle(-Q_O)\psi,\psi\rangle_{L^2(\nu)} = \beta^{-1}\|\nabla_q\psi\|^2_{L^2(\nu)}$, one derives via Cauchy–Schwarz and the Poincaré inequality for ν:

$$\|\nabla_q\psi\|_{L^2(\nu)} \le \beta\sqrt{C_{\mathrm{P}}(\nu)}\,\|(-Q_O)\psi\|_{L^2(\nu)}, \tag{249}$$

and hence $\|\Delta_q\psi\|_{L^2(\nu)} \le \beta(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)})\,\|(-Q_O)\psi\|_{L^2(\nu)}$, where $M_U := \|\nabla U\|_{L^\infty}$.

Combining Steps 1–2 with (247) yields the flux regularity bound $\kappa_{\mathrm{ell}} = c_{d,\beta}\cdot\sqrt{\rho_{\max}/\rho_{\min}}\cdot\beta(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)})$.

E.3. **Perturbation energy bounds (Assumption 3.1.4).** By integration by parts in p, the $L^2(\mu_0)$-adjoint is $\mathcal{V}^\dagger_{\mathrm{pert}} = -F\cdot\nabla_p + \beta(F\cdot p)$, giving the skew/symmetric decomposition:

$$\mathcal{K}_{\mathrm{pert}} = F\cdot\nabla_p - \tfrac{\beta}{2}(F\cdot p), \qquad \mathcal{S}_{\mathrm{pert}} = \tfrac{\beta}{2}(F\cdot p). \tag{250}$$

Since $\|\nabla_p f\| \le \sqrt{\beta}\,\|(-\mathcal{R})^{1/2}f\|$, the bounds reduce to controlling the multiplier $\|pf\|$ by $\|f\|$ and $\|\nabla_p f\|$. A Gaussian integration-by-parts argument with Young's inequality gives the explicit p-multiplier bound:

$$\||p|f\|_{L^2(\mu_0)} \le \sqrt{2d/\beta}\,\|f\|_{L^2(\mu_0)} + (2/\beta)\,\|\nabla_p f\|_{L^2(\mu_0)}. \tag{251}$$

Combining with $\|F\|_{L^\infty} < \infty$ yields Assumption 3.1.4(a)–(b) with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$. □

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